Development of the Vorticity Equation in Duct Flows

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1 Introduction

Vorticity is a key quantity in fluid dynamics, especially in complex flow geometries. This note analyzes the development and evolution of vorticity in two common situations: a curved duct (secondary vorticity generation) and an expanding duct (axial vorticity diffusion).

We start from the Navier-Stokes equations for incompressible flow and derive the vorticity equation by applying the curl operator.

2 Governing Equations

2.1 Navier-Stokes for Incompressible Flow

$$\frac{D\vec{u}}{Dt} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\vec{u}, \quad \text{with} \quad \nabla \cdot \vec{u} = 0 \tag{1}$$

2.2 Vorticity Equation

Taking the curl of the Navier-Stokes equations, we obtain:

$$\frac{D\vec{\omega}}{Dt} = (\vec{\omega} \cdot \nabla)\vec{u} - (\vec{u} \cdot \nabla)\vec{\omega} + \nu\nabla^2\vec{\omega} \tag{2}$$

For incompressible flow, this simplifies to:

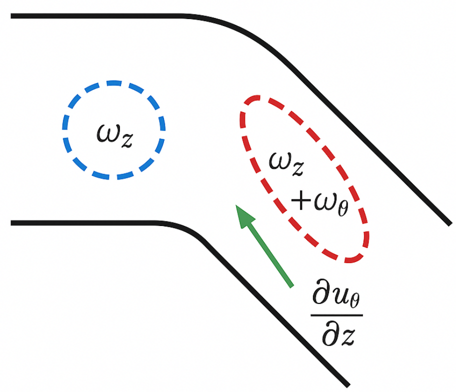
$$\frac{D\vec{\omega}}{Dt} = (\vec{\omega} \cdot \nabla)\vec{u} + \nu\nabla^2\vec{\omega} \tag{3}$$

Where:

- $(\vec{\omega} \cdot \nabla)\vec{u}$: Describes the stretching and tilting of vorticity. *Stretching* occurs when velocity gradients increase the magnitude of vorticity along its direction, while *tilting* refers to the reorientation of vorticity due to transverse velocity gradients.

- $\nu\nabla^2\vec{\omega}$: Represents the viscous diffusion of vorticity, smoothing and spreading it out over time due to the fluid's viscosity.

3 Case 1: 45° Curved Duct



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3.1 Assumptions

- Incompressible, laminar flow ($Re < 2300$)
- Newtonian fluid (shear stress proportional to velocity gradient, constant viscosity)
- Axial symmetry in the duct
- Smooth and steady curvature
- Negligible gravity effects

3.2 Configuration

Due to the duct's curvature, fluid particles near the outer wall must travel a longer path than those near the inner wall. To conserve flow rate, velocity increases near the outer wall and decreases near the inner wall, creating a velocity gradient.

Dominant term:

$$(\vec{\omega} \cdot \nabla)\vec{u} \quad \text{(tilting and stretching)}$$

Justification:

- The curved path induces an azimuthal velocity component u_θ that varies in the axial (z) direction.
- This leads to a tilting/stretching effect:

$$(\vec{\omega} \cdot \nabla)\vec{u} \approx \omega_z \frac{\partial u_\theta}{\partial z}$$

- Axial vorticity is converted into azimuthal vorticity, generating secondary flow structures (e.g., Dean vortices).
- While diffusion is present, it's secondary to the tilting/stretching effects.

Conclusion: In curved geometries, curvature induces stretching and tilting mechanisms that dominate vorticity evolution.

3.3 Mathematical Development

In cylindrical coordinates (r, θ, z), assuming initial axial vorticity ω_z dominates:

$$(\vec{\omega} \cdot \nabla)\vec{u} = \omega_z \left(\frac{\partial u_r}{\partial z}\hat{e}_r + \frac{\partial u_\theta}{\partial z}\hat{e}_\theta + \frac{\partial u_z}{\partial z}\hat{e}_z \right) \tag{4}$$

Since:

$$\frac{\partial u_r}{\partial z} \ll \frac{\partial u_\theta}{\partial z}, \quad \frac{\partial u_z}{\partial z} \approx 0$$

Then the dominant term is:

$$(\vec{\omega} \cdot \nabla)\vec{u} \approx \omega_z \frac{\partial u_\theta}{\partial z}\hat{e}_\theta \tag{5}$$

3.4 Temporal Evolution of Vorticity

Approximating the gradient as:

$$\frac{\partial u_\theta}{\partial z} \approx \frac{\Omega \Delta r}{L_c} \tag{6}$$

Then:

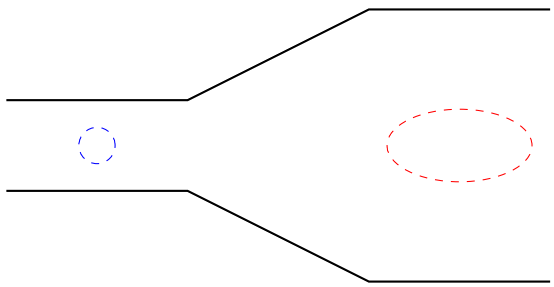
$$\frac{D\omega_\theta}{Dt} \approx \omega_z \frac{\Omega \Delta r}{L_c} - \nu \left(\frac{\omega_\theta}{r^2} \right) \tag{7}$$

- **Initial state:** Vorticity mainly axial, forming a symmetric vortex ring.
- **Entering the curve:** $\partial u_\theta / \partial z$ becomes nonzero due to curvature.
- **Result:** Axial vorticity is tilted into the azimuthal direction, generating new vorticity component ω_θ.
- The vortex ring deforms, becoming elliptical and helicoidal, forming secondary vortices.

Conclusion: Curvature creates ω_θ through stretching of axial vorticity, producing secondary vortex structures.

4 Case 2: Expanding Duct

4.1 Assumptions



- Gradual expansion of cross-sectional area $A(z)$
- Flow rate conserved: $Q = A(z)u_z(z)$
- Initially axial vorticity: $\vec{\omega}(0) = \omega_0 \hat{e}_z$
- Incompressible, laminar flow

Dominant term:

$$\nu \nabla^2 \vec{\omega} \quad \text{(viscous diffusion)}$$

Justification:

- No curvature or velocity gradients that would induce tilting or stretching.
- As the duct expands, axial velocity decreases, and no new vorticity is generated.
- Vorticity spreads radially due to viscosity:

$$\frac{\partial \omega_z}{\partial t} \approx \nu \nabla^2 \omega_z$$

Conclusion: In expanding ducts, vorticity evolution is dominated by viscous diffusion in the absence of stretching or tilting mechanisms.

4.2 Vorticity Evolution

Simplified cylindrical form:

$$\frac{\partial \omega_z}{\partial t} + u_z \frac{\partial \omega_z}{\partial z} = \frac{\nu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \omega_z}{\partial r} \right) \tag{8}$$

Note: $\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}$
Self-similar solutions:

$$\omega_z(r,t) = \frac{\Gamma_0}{4\pi\nu t} \exp\left(-\frac{r^2}{4\nu t}\right) \tag{9}$$

$$R(t) = \sqrt{R_0^2 + 4\nu t} \tag{10}$$

4.3 Circulation Conservation

$$\Gamma = \int_0^{R(z)} \omega_z \cdot 2\pi r \, dr = \Gamma_0 = \text{const} \tag{11}$$

- **Initial state:** Axial vorticity concentrated near the centerline.
- **Effect of expansion:** Larger cross-section reduces velocity, promoting radial vorticity diffusion.
- **Result:** The vortex ring spreads radially, peak vorticity drops, but total circulation is conserved.

Conclusion: Expansion causes axial vorticity to diffuse, broadening its profile while preserving total circulation.

General Comparison

Characteristic	**Curved Duct**	**Expanding Duct**
Vortex ring deformation	Stretching + Tilting	Radial expansion
New vorticity components	Yes ($\omega_\theta \neq 0$)	No
Dominant mechanism	Tilting / Stretching	Viscous diffusion
Vorticity magnitude	May increase locally	Decreases globally
Total circulation	Conserved	Conserved

Table 1: Comparison of vortex ring behavior